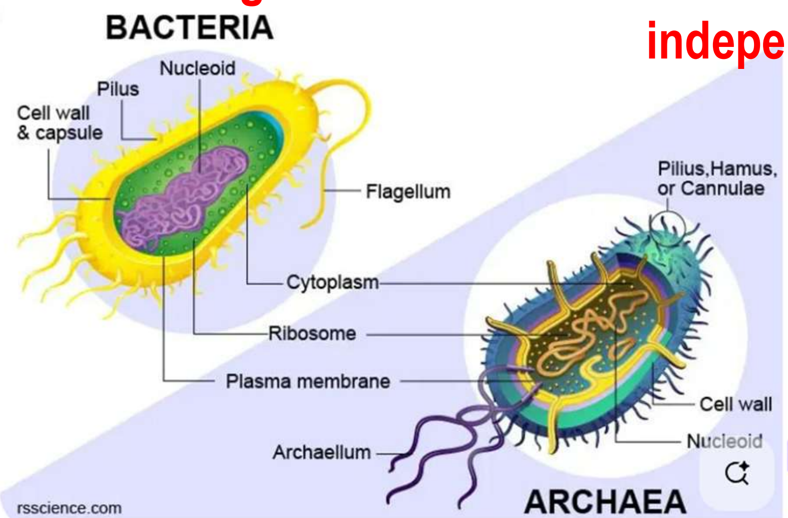
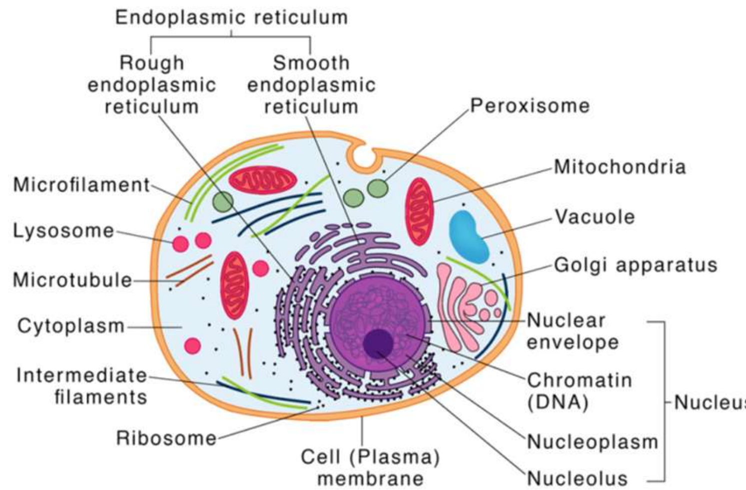
EES 301: Global biogeochemical cycles and Climate change

Evolution of diversity

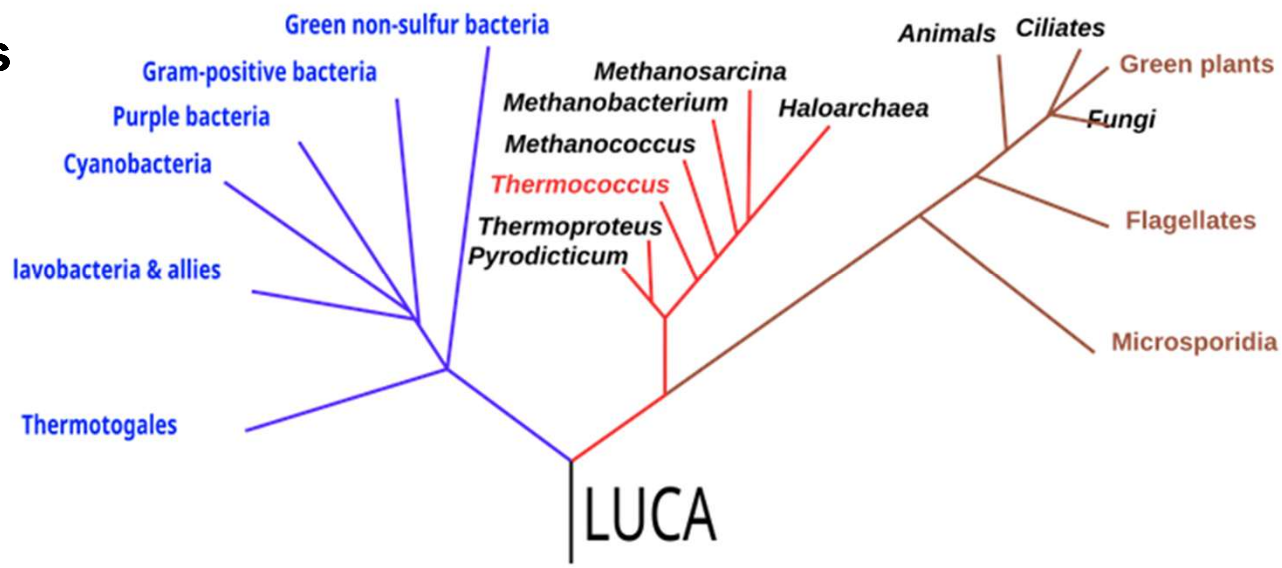
- The main difference between archea and bacteria are differences in the structure of the cell membrane and metabolism. Methanogens, Thermophiles, and Halophiles are archaea
- The evolution of the Eukaryotic cell 2.7 to 1.8 Ga ago presents a milestone in the evolution of life. Protecting the genetic code with an extra membrane protected it from chemical reactions.
- Cell organelles like mitochondria may have originated as independent organisms making the evolution of Eukarya an early case of endosymbiosis



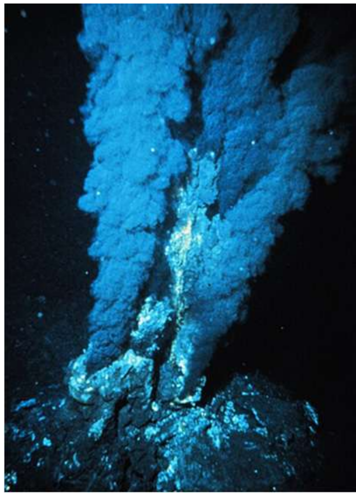


Bacteria Archaea Eukarya

Biological evolution is a change over time in the genetic composition of members of a population of reproducing organisms



Evolution of the global Sulfur cycle



- Many metabolic pathways related to the global sulfur cycle are old.
- The use of sulfur dioxide SO_2 (g) and the products formed when SO_2 dissolves in water HSO_3^- & SO_3^{2-} predate the evolution of photosynthesis as volcanoes & deep sea vents provide SO_2. Volcanoes & SO_2 photolysis also provided H_2S so sulfur oxidation is equally ancient.

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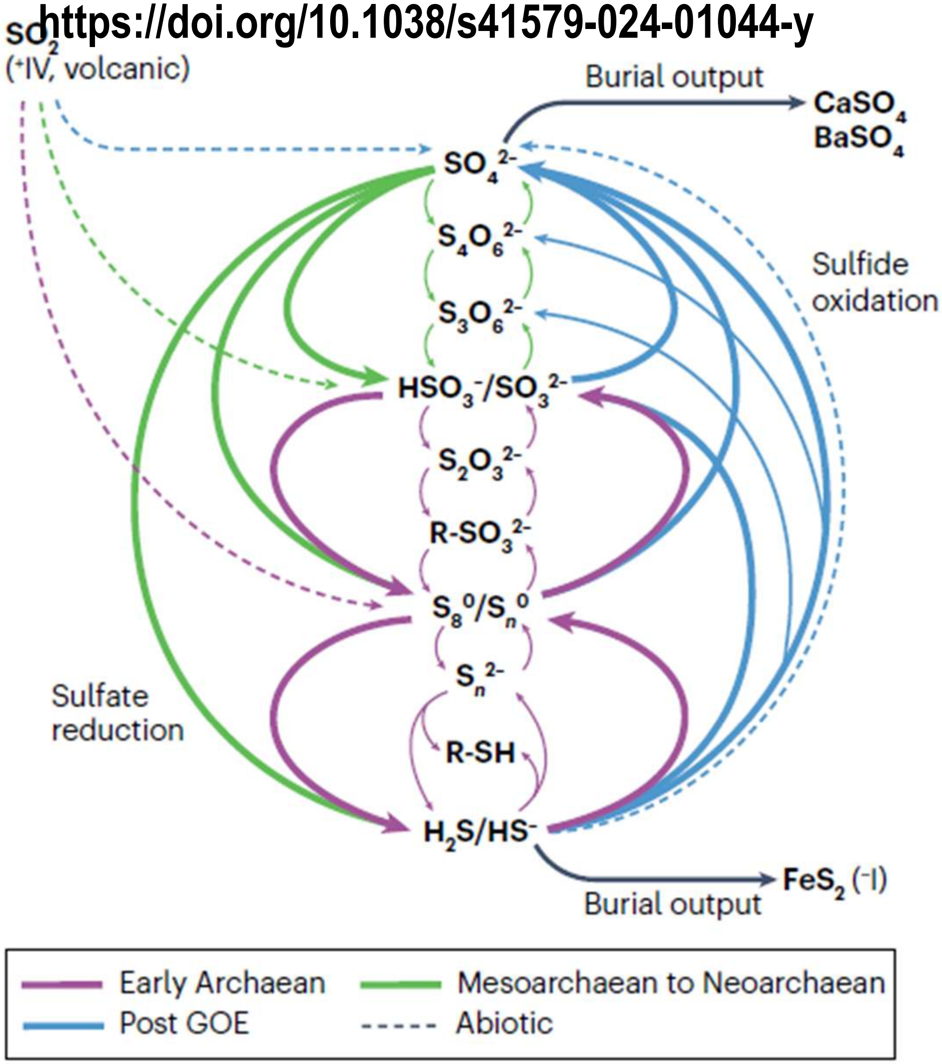
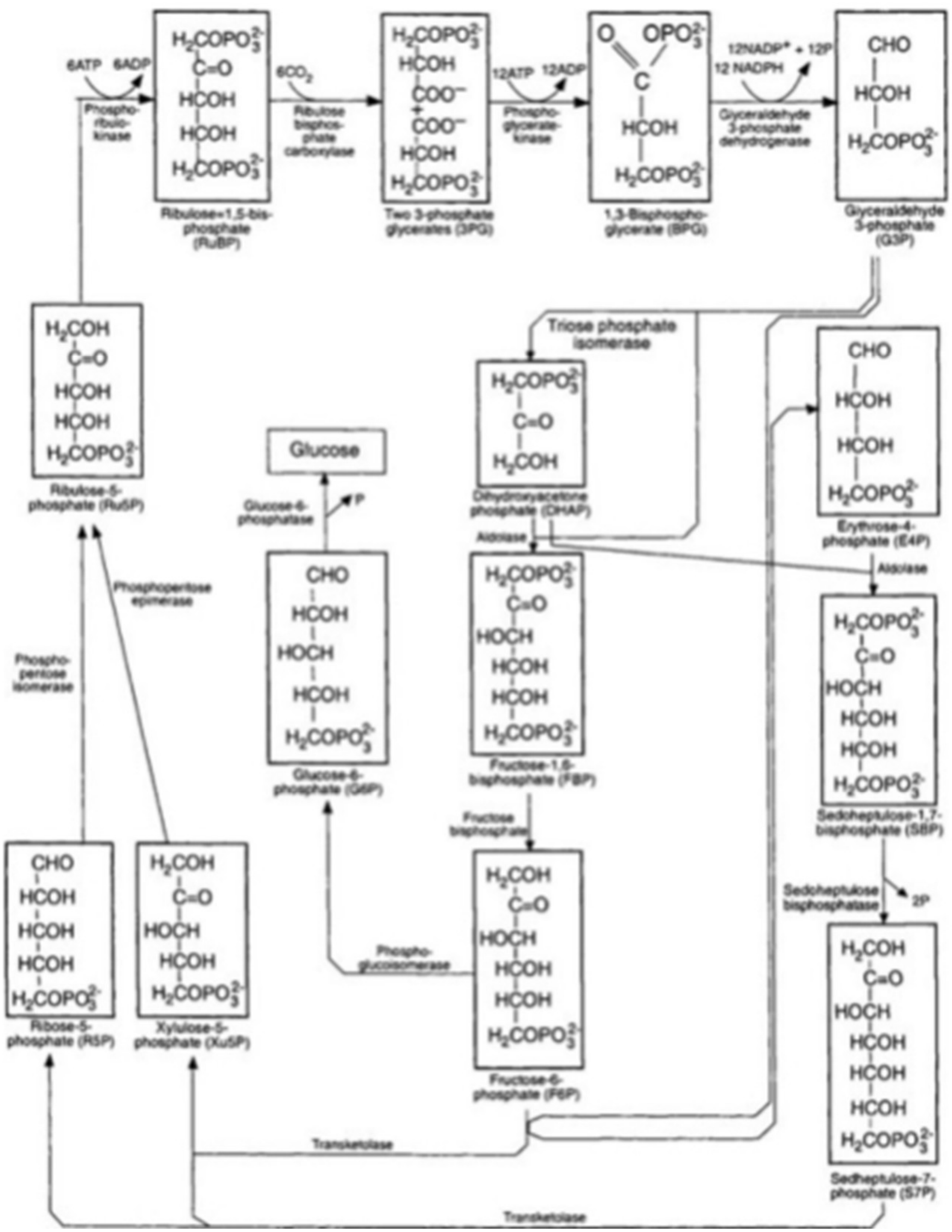


Fig. 4 | The biogeochemical sulfur cycle on Earth. The different metabolic processes involved in the sulfur cycle and the geological intervals within which those metabolisms first expanded in terms of biogeochemical significance. Sulfur is notable for its large range of oxidation states. Sulfate reduction (specifically microbial sulfate reduction) is the process by which sulfate (SO_4^{2-}) is converted to sulfide during the anaerobic respiration of organic matter. Resultant hydrogen sulfide (H_2S) can be an electron source for anoxygenic phototrophs and chemotrophs that oxidize it using various electron acceptors (for example, molecular oxygen (O_2) and nitrate). Sulfide oxidation car proceed directly to sulfate, although this more often occurs via intermediates such as polysulfide (S_n^{2-}), elemental sulfur (S_n^0), thionates (S_nO_m), sulfite (SO_3^{2-}), thiols (R-SH) and sulfonates ($R\text{-}SO_3^{2-}$). All these intermediates participate in bioti and abiotic oxidation, reduction and disproportionation reactions. Pathways involving intermediate valence sulfur species are indicated by thin arrows. Burial output arrows signify that sulfide can react with iron to form sulfide minerals such as pyrite (FeS_2), whereas sulfate can react with calcium or barium to form minerals such as gypsum and anhydrite ($CaSO_4 \pm 2H_2O$) or barite ($BaSO_4$), respectively. Sulfur dioxide (SO_2) gas is primarily produced by volcanoes and can be converted into bioavailable forms through gas-phase and aqueous-phase reactions in the ocean, in the atmosphere and on land. Dashed arrows refer to pathways that do not require life. Arrows are colour-coded according to the geologic interval within which a given metabolic pathway became ecologically significant on a global scale. The Great Oxygenation Event (GOE) spurred the global-scale relevance of the full oxidative suite of metabolisms.

Calvin-Benson Cycle



Photosynthesis

- Autotrophic organisms derive their carbon from CO_2. They use light or inorganic chemicals as energy source.
- The first autotrophic organisms were chemoautotrophic organisms but once photosynthesis evolved, autotrophic organisms performing photosynthesis changed the history of Earth and caused the oxygen holocaust, the first major mass extinction even.
- Plants algae and cyanobacteria fix CO_2. through the Calvin-Benson cycle $CO_2 + H_2O \rightarrow (CH_2O)_n + O_2$
- Purple and green sulfur bacteria perform anoxic photosynthesis

$$CO_2 + H_2S \rightarrow (CH_2O)_n + S^0$$

FIG. 3-5 The Calvin–Benson cycle. This is the series of reactions by which CO_2 is fixed into organic material.

Energy metabolism – aerobic respiration

- In aerobic respiration enzymes remove electrons from organic compounds and pass them through a chain of carriers including flavoproteins and cytochromes located in intracellular membranes until they are finally used to reduce oxygen to produce water.
- In this process ATP is produced by an enzyme called ATPase that is located in the cell membrane and the process is driven by a proton gradient across the membrane.
- Aerobic respiration is very energy efficient. One molecule of glucose produces 2 ATP molecules through lactic acid fermentation but 36 ATP molecules through aerobic respiration.
- The interesting thing about aerobic respiration is that its molecular pathways evolved under anoxic conditions. The electron transport chain is derived from modular enzyme complexes originally evolved for methanogenesis, iron oxidation, and anoxygenic photosynthesis. Membrane bound APTases key to aerobic respriation are also involved when nitrifying bacteria oxidize ammonia to nitrite or nitrite to nitrate, sulfur bacteria oxidize hydrogen sulfide to or sulfur to sulfate, iron bacteria oxidize reduced ferrous iron to form hematite and hydrogen bacteria use hydrogen gas as energy source.

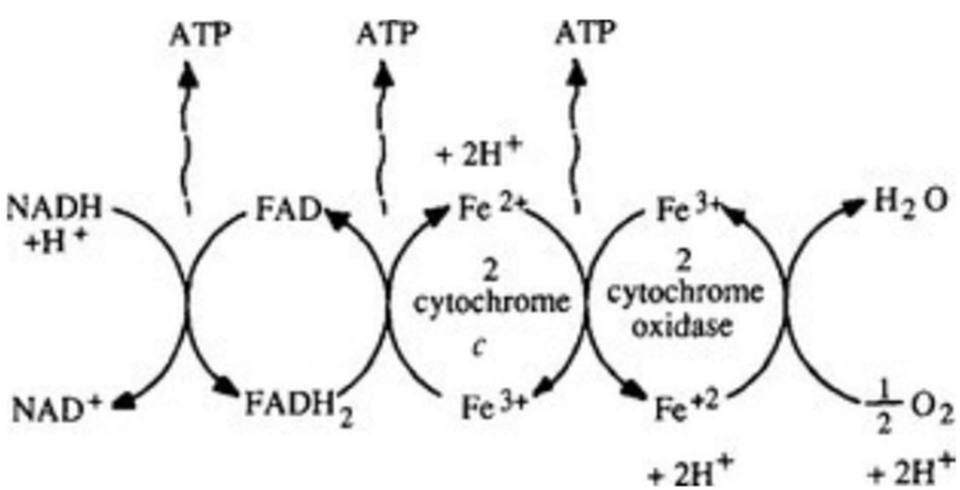


FIG. 3-4 Electron transport process schematic, showing coupled series of oxidation–reduction reactions that terminate with the reduction of molecular oxygen to water. The three molecules of ATP shown are generated by an enzyme called ATPase which is located in the cell membrane and forms ATP from a proton gradient created across the membrane.

Evolution of Photosynthesis

- The evolution of photosynthesis ~ 3.5 Ga ago presents another milestone in the evolution of life. It resulted into an energy bonanza as microbes became independent of geological energy sources fueled by the plate tectonic system and relied on the more abundant sunlight
- It did not directly result in the rise of free oxygen. Instead a co-evolution of eukaryotic cells containing mitochondria that performed aerobic respiration ~2.7 to 1.8 Ga ago meant that the biosphere was the first oxygen sink.
- Iron became the next big oxygen sink. Banded iron formations formed.

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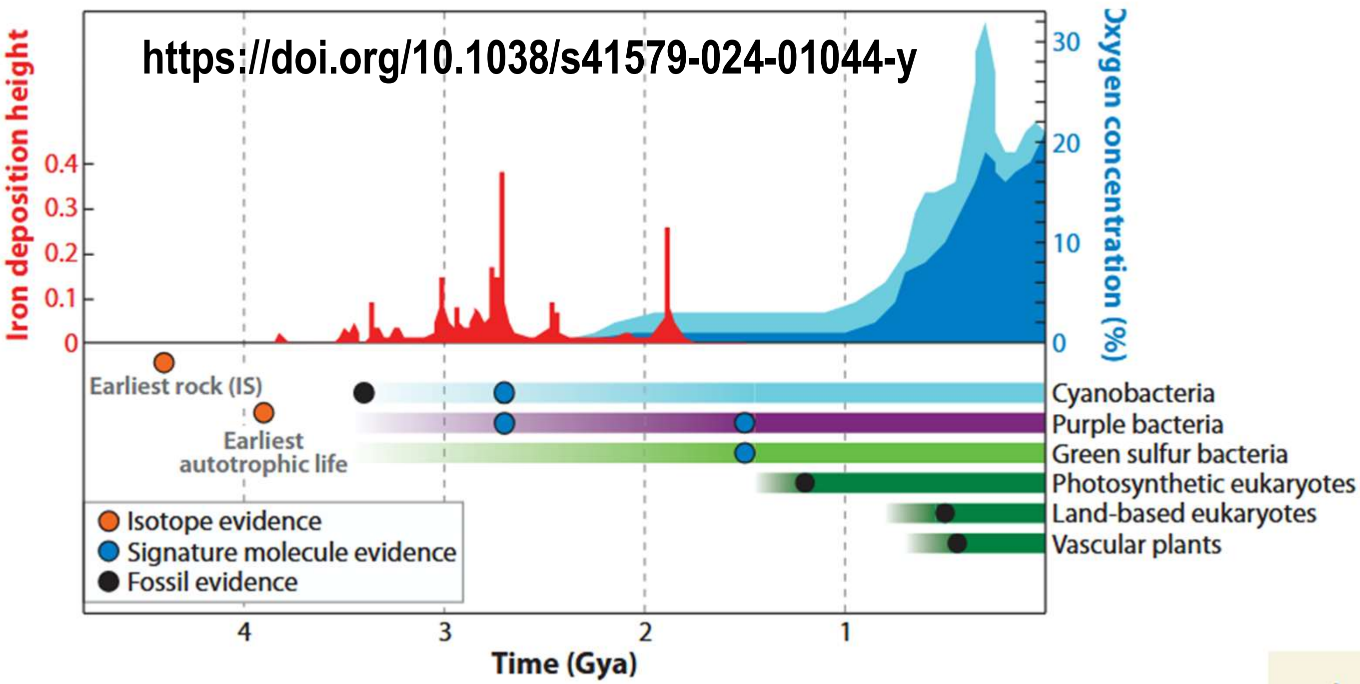
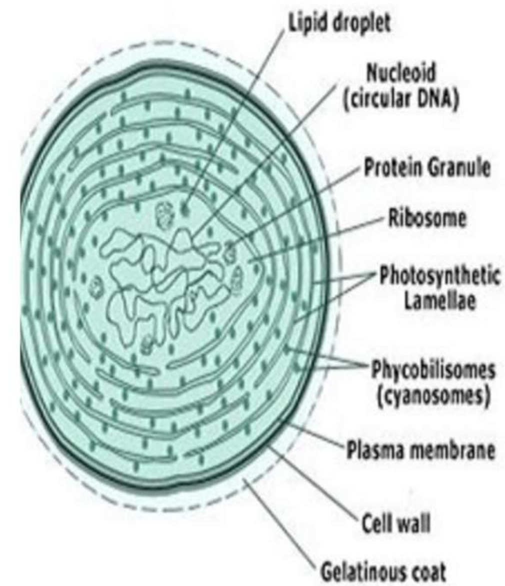


Figure 2

Evolution of life and photosynthesis in geological context, highlighting the emergence of groups of photosynthetic organisms. Minimum and maximum estimates for oxygen concentration are indicated by dark blue and light blue areas, respectively. Oxygen concentration data from Reference 80; banded iron formations data from Reference 87.

Cyanobacteria



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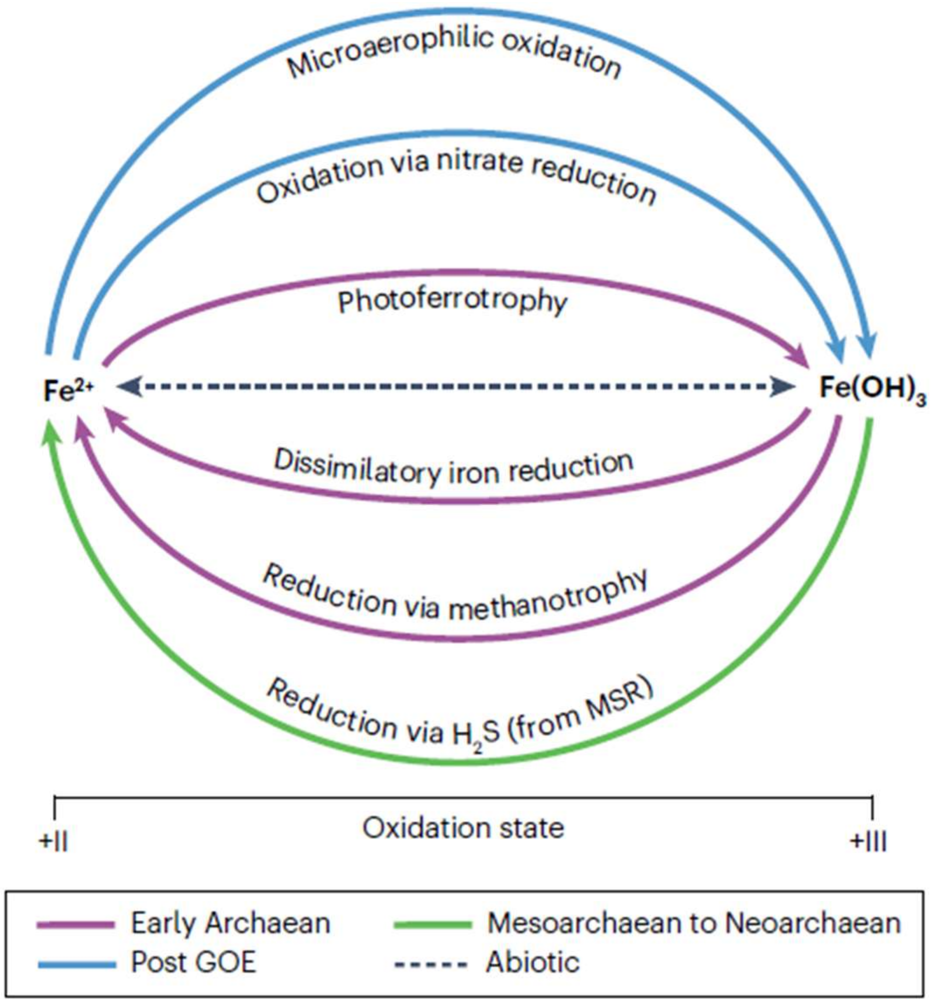
Annu. Rev. Plant Biol. 2011. 62:515–48

Evolution of the global Iron cycle

- Many metabolic pathways related to iron cycling are old and likely evolved before photosynthesis. Both the photoferrotrophic oxidation and the microbial reduction are believed to play a role in the deposition of banded iron formations (BIF)



Banded Iron Formation (BIF) from Australia

Fig. 3 | The biogeochemical iron cycle on Earth. A simple representation of the different iron metabolic processes and the geological intervals within which those metabolisms first expanded in terms of biogeochemical significance. Arrows are colour-coded according to the geologic interval within which a given metabolic pathway became ecologically significant on a global scale. Although oceanic iron is present in only two oxidation states – ferrous iron (Fe^{2+}) and ferric hydroxide ($Fe(OH)_3$) – a wide variety of metabolisms are known to leverage this limited range. Iron is oxidized by chemolithoautotrophic bacteria at low oxygen concentrations (microaerophilic oxidation). Such oxidation can also occur using nitrate as the electron acceptor (oxidation via nitrate reduction). Photoferrotrophy is the oxidation of reduced iron via anoxygenic photosynthesis. Resultant oxidized iron from any of the above pathways can be reduced during the anaerobic respiration of biomass (dissimilatory iron reduction (DIR)). Reduction of oxidized iron can also occur using methane (reduction via methanotrophy) or sulfide (reduction via hydrogen sulfide (H_2S)) as the electron donor. In this way, shifts in the dominant pathways of iron cycling are intimately linked to selective environmental and/or evolutionary pressures, including the rise of oxygenic photosynthesis approximately 3.0 billion years ago and subsequent accumulations of nitrate and H_2S in the Proterozoic ocean. GOE, Great Oxidation Event; MSR, microbial sulfate reduction.

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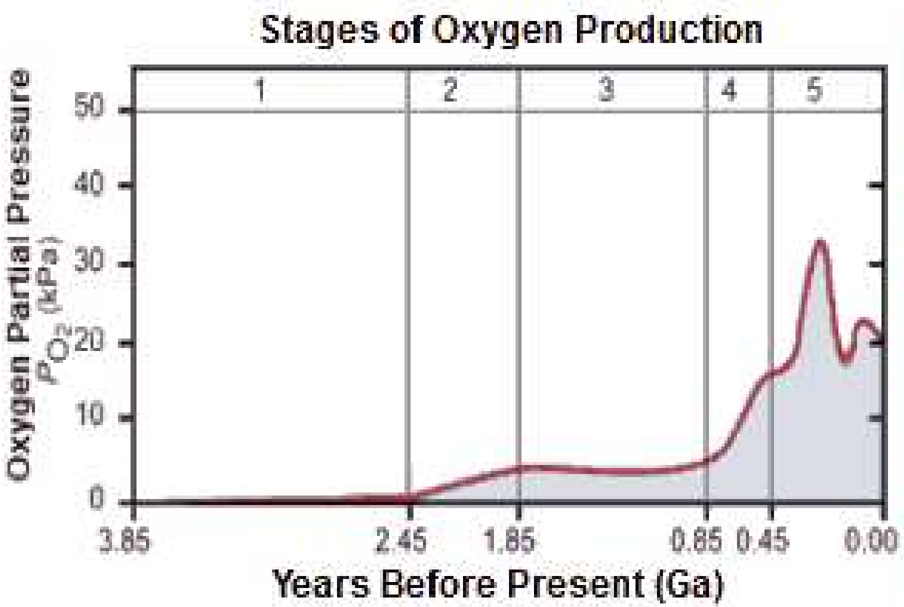
The first mass extinction was triggered by oxygen



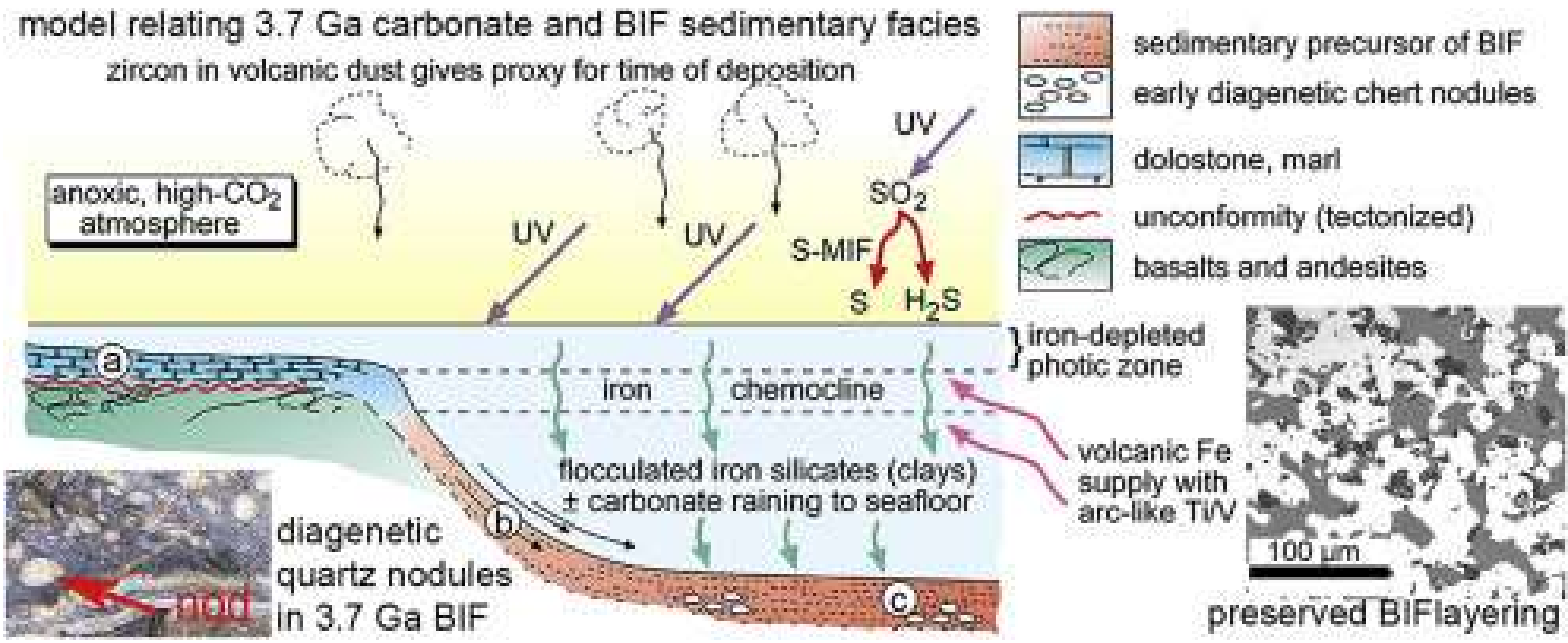
Banded Iron Formation (BIF) from Australia

The red bands are hematite an iron III mineral

Stomatolites are created by photosynthetic cyanobacteria, the oldest are 3.5 Ga old

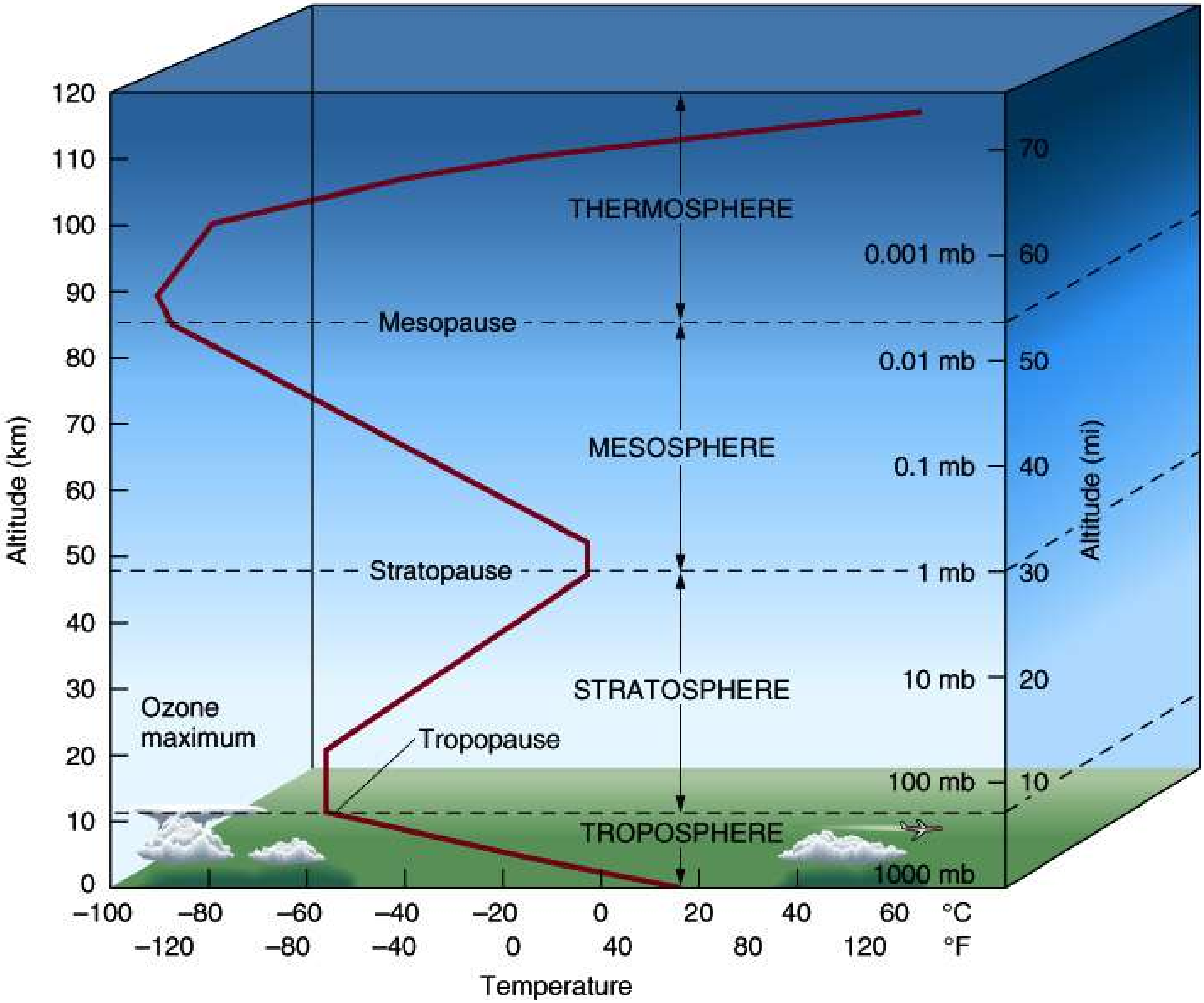
It resulted in the extinction of many forms of chemoautotrophs to whom oxygen is toxic. Some chemoautotrophs still reside in extreme habitats e.g. around deep see volcanoes as extremophiles.

The oxidation of the atmosphere allowed the formation of the stratospheric ozone layer

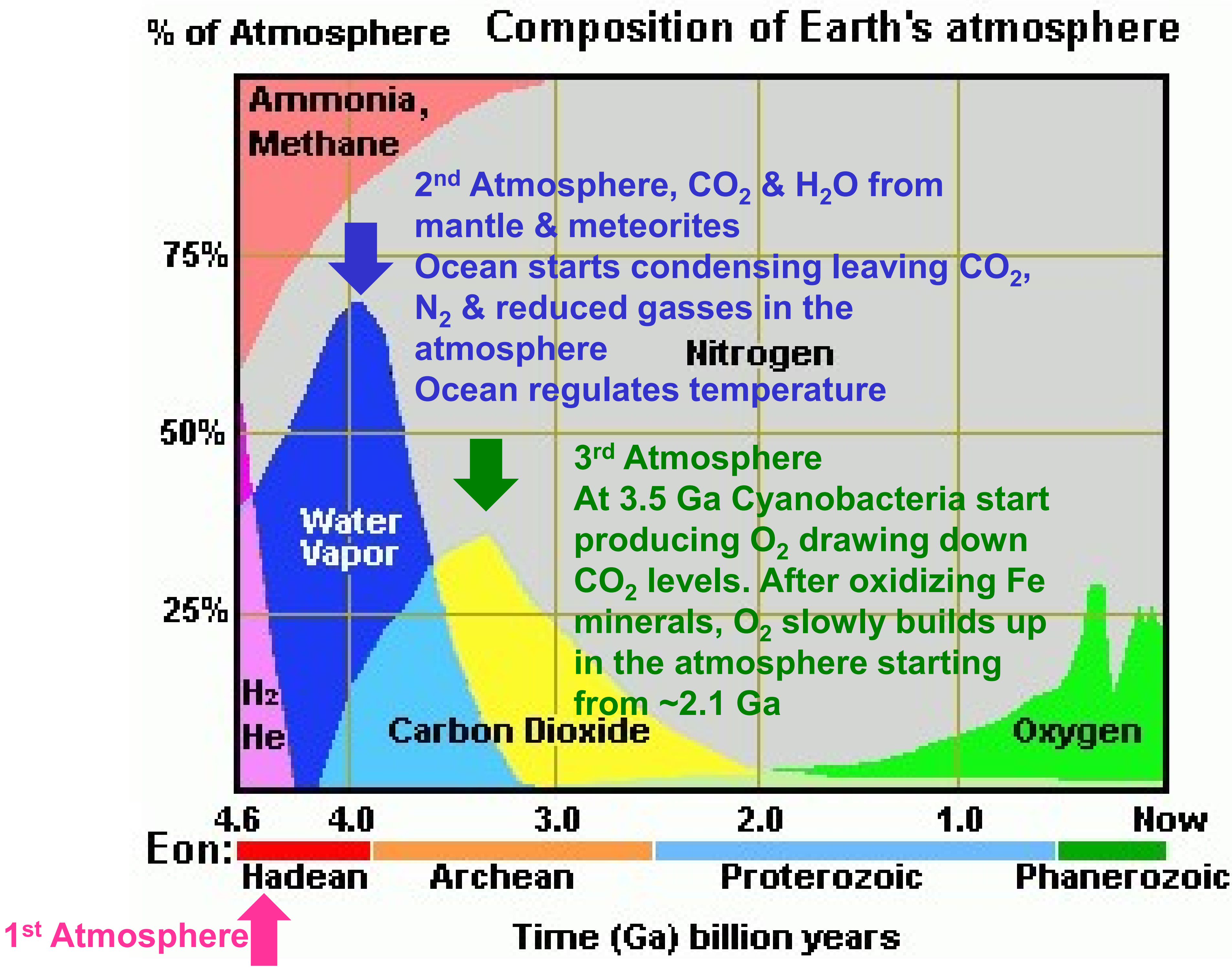


THE STRATOSPHERIC OZONE LAYER PROTECTS LIFE ON LAND FROM HARMFUL UV RAYS

The atmosphere today



Good ozone



% of Atmosphere
Composition of Earth's atmosphere
Ammonia, Methane
2nd Atmosphere, CO_2 & H_2O from mantle & meteorites
Ocean starts condensing leaving CO_2, N_2 & reduced gasses in the atmosphere
Ocean regulates temperature
Nitrogen
75%
50%
25%
Water Vapor
3rd Atmosphere
At 3.5 Ga Cyanobacteria start producing O_2 drawing down CO_2 levels. After oxidizing Fe minerals, O_2 slowly builds up in the atmosphere starting from ~2.1 Ga
H_2
He
Carbon Dioxide
Oxygen
4.6
4.0
3.0
2.0
1.0
Now
Eon:
Hadean
Archean
Proterozoic
Phanerozoic
1st Atmosphere
Time (Ga) billion years

Evolution of the global Nitrogen cycle

- **Most metabolic processes related to the nitrogen cycling evolved late in the Archean or after the great oxidation event.**
- **Only nitrogen fixation appears to have evolved in the Early Archean.**

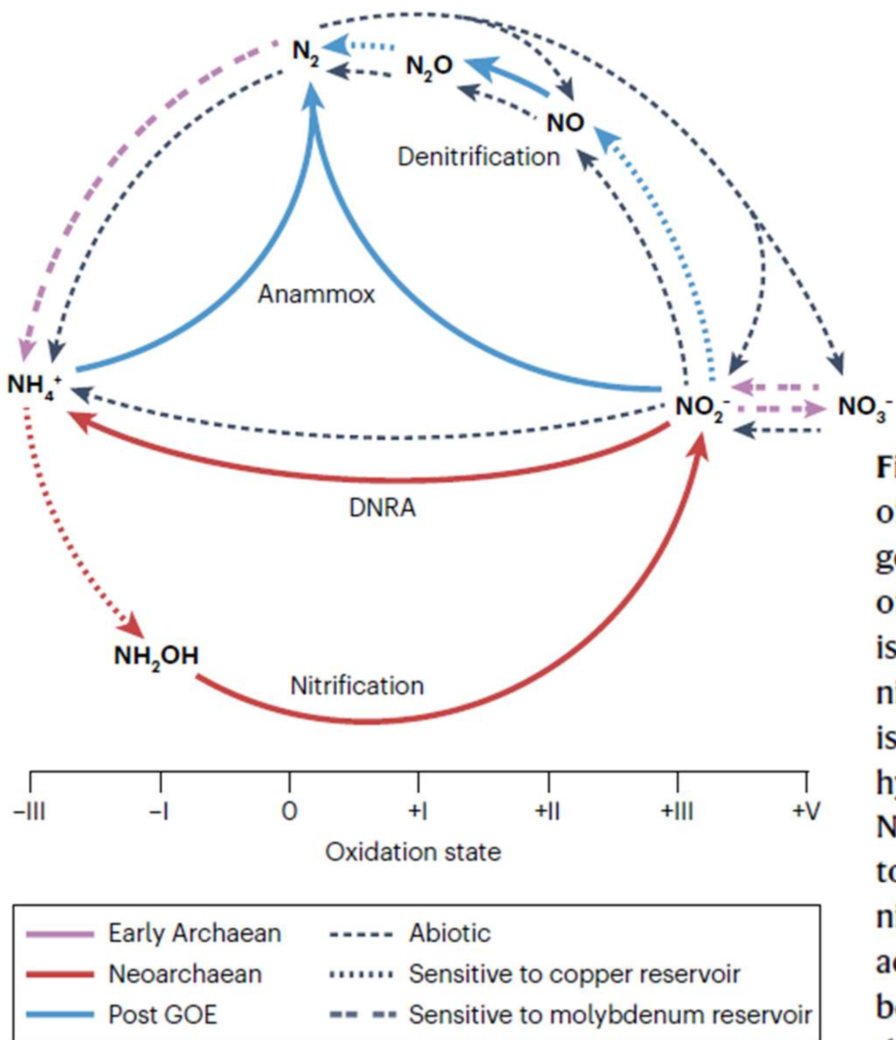


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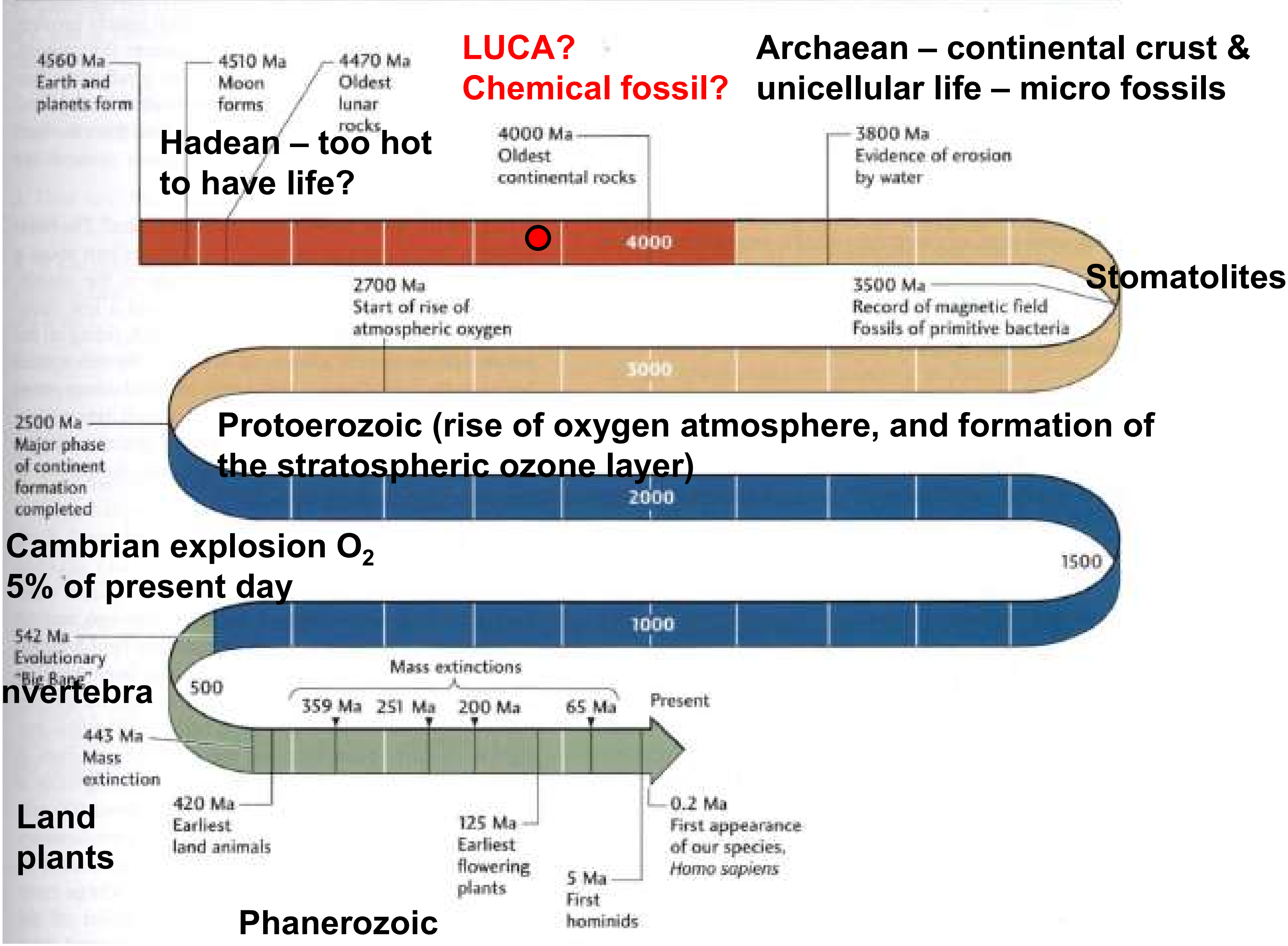
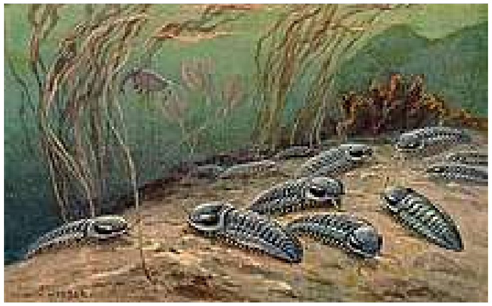
Fig. 2 | The biogeochemical nitrogen cycle on Earth. A simple representation of the interactions among different nitrogen metabolic processes and the geological intervals within which those metabolisms first expanded in terms of biogeochemical significance. Anaerobic ammonium oxidation (anammox) is a process by which microorganisms oxidize ammonium (NH_4^+) to molecular nitrogen (N_2) gas by utilizing nitrite (NO_2^-) as an electron acceptor. Nitrification is the enzymatic oxidation of ammonium to nitrite and nitrate (NO_3^-) via hydroxylamine (NH_2OH). Denitrification is the reduction of oxidized nitrogen to N_2 under low molecular oxygen (O_2) conditions. Dissimilatory nitrate reduction to ammonium (DNRA) is also an anaerobic reduction process, but it retains fixed nitrogen as ammonium rather than generating N_2. Arrows are colour-coded according to the geologic interval within which a given metabolic pathway became ecologically significant on a global scale. Nitrogen cycling is notably dependent on the oceanic molybdenum and copper reservoirs for key enzymatic cofactors, and variations in the sizes of those reservoirs may have set the tempo of a given metabolism's proliferation through time. GOE, Great Oxygenation Event.

Key Figure 1.13 The geologic record shows evidence of global geosystems operating in deep time.



Paleozoic Era

Cambrian Period



Almost all marine phyla evolved in this period.

Ordovician Period

Evolution of primitive fish, cephalopods, and coral

Silurian Period

Fully terrestrial life evolved, including early arachnids, fungi, and centipedes.

Devonian Period

On land, plant groups diversified incredibly in an event known as the Devonian Explosion when plants made lignin allowing taller growth and vascular tissue: the first trees evolved,

Carboniferous Period (359 to 299 million years ago)

Average global temperatures were high at the start and low at the end; the early Carboniferous averaged at about 20 degrees Celsius (but cooled to 10 °C during the Middle Carboniferous). Tropical swamps dominated the Earth, and the lignin stiffened trees grew to greater heights and number. As the bacteria and fungi capable of eating the lignin had not yet evolved, their remains were left buried, which created much of the carbon that became the coal deposits of today

Permian Period

The Permian spanned from 299 to 252 million years ago and was the last period of the Paleozoic Era. At the beginning of this period, all continents joined together to form the supercontinent Pangaea

Permian–Triassic extinction event

- Around 81% of all marine species and 70% of terrestrial vertebrate species became extinct. It was the largest known mass extinction of insects. Some 57% of all biological families and 83% of all genera became extinct.
- The eruption of the Siberian Traps coincide with the event. Evidence for widespread ocean acidification, ocean anoxia (severe deficiency of oxygen) and euxinia (presence of hydrogen sulfide) is found from the Late Permian to the Early Triassic. CO_2 levels topped at 2000 ppm.

Mesozoic

70% of oil deposits existing today were formed in the Mesozoic age (252 to 66 million years ago), 20% were formed in the Cenozoic age (65 million years ago to present), and only 10% were formed in the Paleozoic age (541 to 252 million years ago).

The Mesozoic was a time of significant tectonic, climate, and evolutionary activity. The era witnessed the gradual rifting of the supercontinent Pangaea into separate landmasses that would move into their current positions during the next era. The climate of the Mesozoic was varied, alternating between warming and cooling periods. Overall, however, the Earth was hotter than it is today. Dinosaurs first appeared in the Mid-Triassic, and became the dominant terrestrial vertebrates in the Late Triassic or Early Jurassic, occupying this position for about 150 or 135 million years until their demise at the end of the Cretaceous. Birds first appeared in the Jurassic (however, true toothless birds appeared first in the Cretaceous), having evolved from a branch of theropod dinosaurs. The first mammals also appeared during the Mesozoic, but would remain small—less than 15 kg (33 lb)—until the Cenozoic. The flowering plants (angiosperms) appeared in the Early Cretaceous and would rapidly diversify throughout the period, replacing conifers and other gymnosperms as the dominant group of plants.

Cretaceous–Paleogene extinction event

The Cretaceous–Paleogene (K–Pg) extinction event, also known as the Cretaceous–Tertiary (K–T) extinction, was a sudden mass extinction of three-quarters of the plant and animal species on Earth, approximately 66 million years ago. With the exception of some ectothermic species such as the sea turtles and crocodilians, no tetrapods weighing more than 25 kilograms (55 pounds) survived

In the geologic record, the K–Pg event is marked by a thin layer of sediment called the K–Pg boundary, which can be found throughout the world in marine and terrestrial rocks. The boundary clay shows unusually high levels of the metal iridium, which is more common in asteroids than in the Earth's crust.

The Deccan Trapps also erupted at the end of the Cretaceous

Cenozoic

- The Cenozoic is also known as the Age of Mammals because the terrestrial animals that dominated both hemispheres were mammals – the Eutherians (placentals) in the northern hemisphere and the Metatherians (marsupials, now mainly restricted to Australia) in the southern hemisphere. The extinction of many groups allowed mammals and birds to greatly diversify so that large mammals and birds dominated the Earth. The continents also moved into their current positions during this era.

Holocene extinction

- The Holocene extinction, otherwise referred to as the sixth mass extinction or Anthropocene extinction, is an ongoing extinction event of species during the present Holocene epoch (with the more recent time sometimes called Anthropocene) as a result of human activity.
- The included extinctions span numerous families of plants and animals, including mammals, birds, reptiles, amphibians, fishes and invertebrates.
- With widespread degradation of highly biodiverse habitats such as coral reefs and rainforests, as well as other areas, the vast majority of these extinctions are thought to be undocumented, as the species were undiscovered at the time of their extinction, or no one has yet discovered their extinction. The current rate of extinction of species is estimated at 100 to 1,000 times higher than natural background extinction rates.
- In *The Future of Life* (2002), Edward Osborne Wilson of Harvard calculated that, if the current rate of human disruption of the biosphere continues, one-half of Earth's higher lifeforms will be extinct by 2100.
- At present, the rate of extinction of species is estimated at 100 to 1,000 times higher than the background extinction rate, the historically typical rate of extinction (in terms of the natural evolution of the planet);
- The current rate of extinction is 10 to 100 times higher than in any of the previous mass extinctions in the history of Earth.

A new geological Era normally starts with a mass extinction event!

The Anthropocene may have started with the extinction of the ice-age megafauna, was followed by deforestation for agriculture and smelting. However extinction became global and large scale with modern agrochemicals and the expansion of irrigation agriculture using fossil water during the great acceleration – based on this fact the Anthropocene could become an Era

Anthropocene

- The Anthropocene is a proposed geological epoch dating from the commencement of significant human impact on Earth's geology and ecosystems, including, but not limited to, anthropogenic climate change
- As of July 2020, neither the International Commission on Stratigraphy (ICS) nor the International Union of Geological Sciences (IUGS) has officially approved the term as a recognised subdivision of geologic time, although the Anthropocene Working Group (AWG) of the Subcommission on Quaternary Stratigraphy (SQS) of the ICS voted in April 2016 to proceed towards a formal golden spike (GSSP) proposal to define the Anthropocene epoch in the geologic time scale and presented the recommendation to the International Geological Congress in August 2016

Various start dates for the Anthropocene have been proposed, ranging from the beginning of the Agricultural Revolution 12,000–15,000 years ago, to as recent as the 1960s. The ratification process is still ongoing, and thus a date remains to be decided definitively, but the peak in radionuclides fallout consequential to atomic bomb testing during the 1950s has been more favoured than others, locating a possible beginning of the Anthropocene to the detonation of the first atomic bomb in 1945, or the Partial Nuclear Test Ban Treaty in 1963.

PRACTICE QUESTIONS

You can submit the answer to these questions by Tuesday 19th August if you want to get them corrected before the Midsem exam

MCQ style

1) Which of the following is important because of its role in both the movement of continents and the distribution and evolution of species?
   a) Volcanic eruptions
   b) Natural selection
   c) Earthquakes
   d) Plate tectonics
2) What were the earliest forms of life?
   a) Bacteria
   b) Worms
   c) Jellyfish
   d) Plants
3) What molecule, also called the biological blueprint for life, do all living organisms possess?
   a) H2O
   b) DNA
   c) C6H806
   d) CO2

MCQ style

1) **Which of the following best describes the role life itself has had in the history of the biosphere?**
   a) It had little impact on the biosphere until humans emerged and developed land-altering technologies.
   b) It has always worked toward its own continuation and development.
   c) It has altered the atmosphere's composition and influenced the carbon cycle.
   d) It has altered the composition of all the Earth's layers through nutrient cycling.
2) **Around 2.4 billion years ago, the Great Oxidation Event occurred, and many anaerobic respirators died off in an ecological disaster. But it wasn't all bad. What was the major benefit of this event?**
   a) The Earth's temperature dropped
   b) The formation of the ozone layer
   c) A chance for mammals to thrive
   d) The emergence of multicellular organisms
3) **Why was the first mini threshold of life's evolution, photosynthesis, so important?**
   a) It allowed life to spread beyond oceanic vents.
   b) It transformed our atmosphere from oxygen-rich to carbon dioxide-rich.
   c) It helped plants move onto land.
   d) It forced organisms to become multicellular to process more complex energy.

MCQ style

1) What crucial astronomical factor massively influenced the history of the biosphere? Choose the best answer
   a) The impact of Earth's moon on its climate
   b) The release of greenhouse gases due to movement of tectonic plates
   c) The impact of photosynthesizing organisms on our atmosphere
   d) The relationship between the Earth and the Sun
2) How are archaea different from bacteria? Choose 2 answers:
   a) Archaea have membrane-bound organelles.
   b) The chemical composition of the cell wall and cell membrane in archaea are different.
   c) Archaea have more complicated RNA polymerases.
   d) All archaea are found in extreme environments.
3) Which of the following are constituents in all bacteria? Choose 2 answers:
   a) DNA
   b) Chlorophyll
   c) Cell wall
   d) Flagella

MCQ style

1) Which of the following statements are true about photosynthetic and chemosynthetic bacteria? Choose 2 answers:
   a) Both of them can make their own organic food molecules from simple inorganic molecules.
   b) Photosynthetic bacteria are autotrophic while chemosynthetic bacteria are heterotrophic.
   c) Chemosynthetic bacteria can make ATP molecules, but not their own food.
   d) Chemosynthetic bacteria oxidize inorganic compounds for energy
2) What simple organic molecules were likely the precursors to life?
   a) Water and proteins
   b) Prokaryotes and eukaryotes
   c) Oxygen and carbon
   d) Amino acids and nucleotides
3) Where on the Earth did life mostly likely first emerge, where all the ingredients and conditions for life existed?
   a) Volcanic islands
   b) Deep oceanic vents
   c) Tidal pools
   d) Tropical rainforests

Long answer style

1. Why does Earth's sister planet Venus not support life?
2. The Early Earth is thought to have been devoid of oxygen. How did oxygen first appear in the Earth's atmosphere?
3. Which abiotic substances critical to the evolution of life may have been delivered by meteorites and what is the key differences between the biotic and abiotic synthesis of these compounds?
4. What are the key requirements for life that must have been there from the earliest forms?
5. What are anaerobic bacteria and when did they evolve?
6. How does the anaerobic respiration differ from aerobic respiration and when did aerobic respiration evolve?
7. Why was the evolution of photosynthesis an important milestone in the history of Earth and which consequences did it have?
8. What is an autotroph how does it differ from an heterotroph?